Lecture 2 - Data Analysis

ECE 447: Data Analysis and Machine Learning for Engineers

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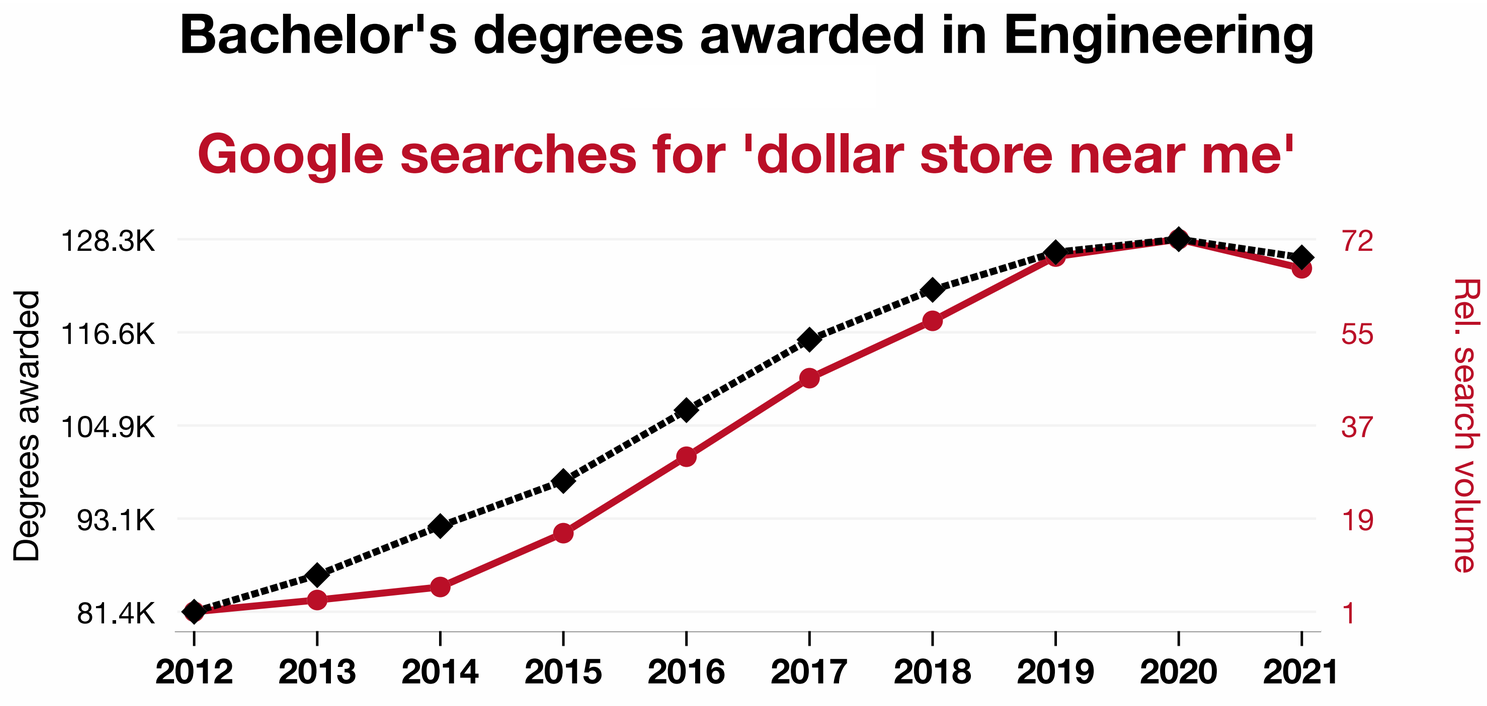
Land Acknowledgment

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

Spurious Correlations *(A correlation that does not imply causation)*



Medical imaging example: ML is susceptible to learn non-medical, irrelevant patterns instead of true underlying disease signs (a shortcut learning) for prediction.

figure from https://www.tylervigen.com/spurious-correlations?page=2

This Lecture Covers ...

- Predictive Analytics and Machine Learning
  - Inductive Bias versus Sample Bias
  - Anything Wrong with ML
  - Lifecycle

- Data Exploration
  - Characteristics
  - Quality
  - Preparation

Predictive Data Analytics

Art of building and using models that make predictions based on patterns extracted from historical data.

— *ML for Predictive Data Analytics*

Applications

- Price prediction
- Dosage prediction
- Risk Assessment
- Likelihood modelling
- Diagnosis
- Document classification

Machine Learning

*An automated process that extracts **patterns** from data. It captures the relationship between features and target ...*

— ML for Predictive Data Analytics

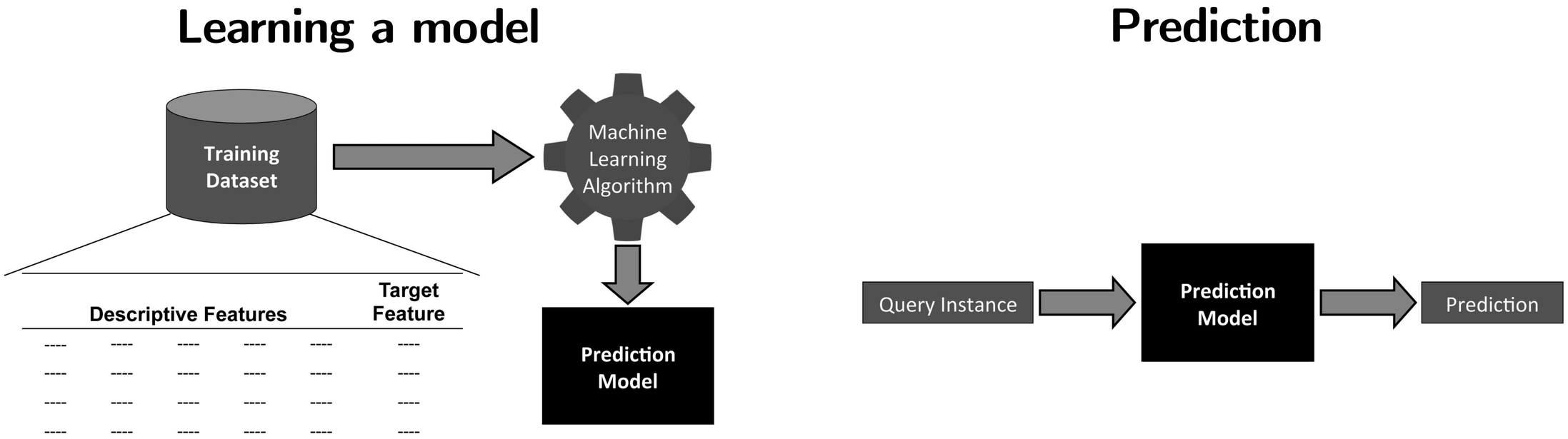


figure from ML for Predictive Data Analytics textbook.

Credit Scoring Dataset Example

Goal: find a model that predicts the outcome (default loan, or repay).

You might be able to find a logic algorithm.

if LOAN-SALARY RATIO < 1.5 **then**
  OUTCOME = *repay*
else if LOAN-SALARY RATIO > 4 **then**
  OUTCOME = *default*
else if AGE < 40 **and** OCCUPATION $=$ *industrial* **then**
  OUTCOME = *default*
else
  OUTCOME = *repay*
end if

ID	AMOUNT	SALARY	LOAN-SALARY RATIO	AGE	OCCUPATION	PROPERTY	TYPE	OUTCOME
1	245,100	66,400	3.69	44	industrial	farm	stb	repay
2	90,600	75,300	1.20	41	industrial	farm	stb	repay
3	195,600	52,100	3.75	37	industrial	farm	ftb	default
4	157,800	67,600	2.33	44	industrial	apartment	ftb	repay
5	150,800	35,800	4.21	39	professional	apartment	stb	default
6	133,000	45,300	2.94	29	industrial	farm	ftb	default
7	193,100	73,200	2.64	38	professional	house	ftb	repay
8	215,000	77,600	2.77	17	professional	farm	ftb	repay
9	83,000	62,500	1.33	30	professional	house	ftb	repay
10	186,100	49,200	3.78	30	industrial	house	ftb	default
11	161,500	53,300	3.03	28	professional	apartment	stb	repay
12	157,400	63,900	2.46	30	professional	farm	stb	repay
13	210,000	54,200	3.87	43	professional	apartment	ftb	repay
14	209,700	53,000	3.96	39	industrial	farm	ftb	default
15	143,200	65,300	2.19	32	industrial	apartment	ftb	default
16	203,000	64,400	3.15	44	industrial	farm	ftb	repay
17	247,800	63,800	3.88	46	industrial	house	stb	repay
18	162,700	77,400	2.10	37	professional	house	ftb	repay
19	213,300	61,100	3.49	21	industrial	apartment	ftb	default
20	284,100	32,300	8.80	51	industrial	farm	ftb	default
21	154,000	48,900	3.15	49	professional	house	stb	repay
22	112,800	79,700	1.42	41	professional	house	ftb	repay
23	252,000	59,700	4.22	27	professional	house	stb	default
24	175,200	39,900	4.39	37	professional	apartment	stb	default
25	149,700	58,600	2.55	35	industrial	farm	stb	default

But, it's almost impossible to construct this model by hand. Hence, with machine learning, we learn a predictive pattern from data.

figure from ML for Predictive Data Analytics textbook.

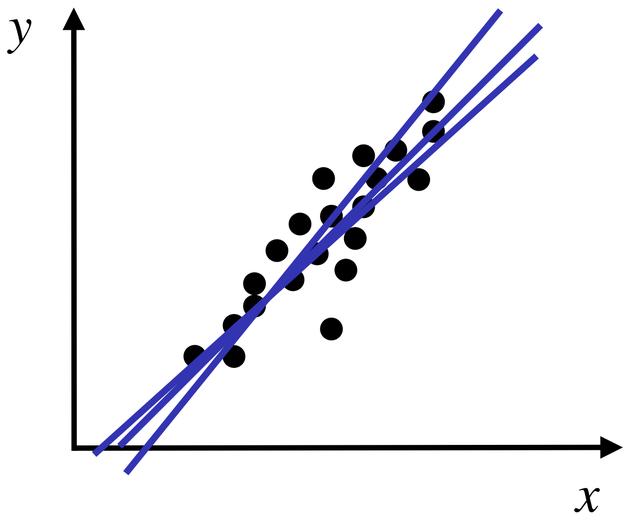
How to Learn? What Are the Challenges?

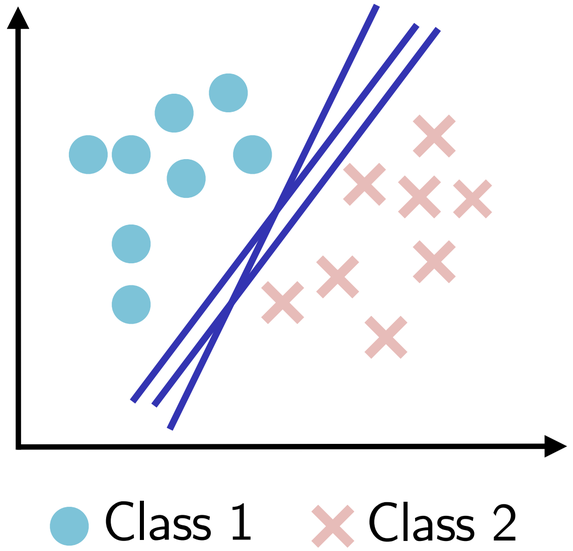
Algorithms work by ***searching through a set of possible prediction models*** *for the model that best captures the relationship between input (descriptive features) and target based on available data.*

— *ML for Predictive Data Analytics*

Challenges

- Data is noisy
- The problem is ill-posed; given the available data, solution may not be unique.



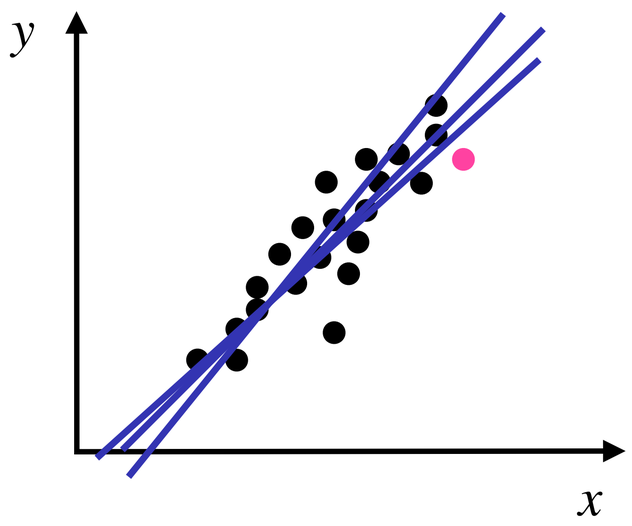


Model Generalization

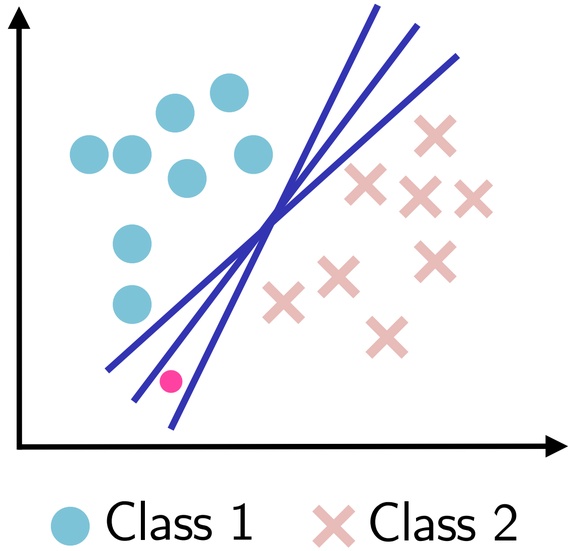
For a prediction model to be useful, it should be able to make predictions on **new** data **not** present in the **training** dataset.

Good prediction (high accuracy) —> we say the model **generalizes** well.

Given new data x •, predict y.



Given new data •, predict the class.



Training, Validation, Test Datasets in Machine Learning

Training Dataset

*The data samples used to **fit** (train) the model.*

Validation Dataset

The data samples used to provide unbiased evaluation of the model (choice of model, tune the model).

Test Dataset

The data samples used to provide an unbiased evaluation of the final model.

figure from https://medium.com/data-science/train-validation-and-test-sets-72cb40cba9e7

How to Choose the Model?

Machine learning methods (models) have distinct characteristics,
i.e., rely on different criteria. This is an ***inductive bias****.*

Inductive bias of the machine learning algorithm:

- **Restriction bias**

  Constrains a set of models that the algorithm considers during the learning process

  e.g., use linear regression, decision trees

- **Preference bias**

  Guides the learning algorithm to prefer certain models over others.

  e.g., order of the linear model, shallow trees

Data Properties

In addition to the **inductive bias**,

we look into properties of **the training dataset**.

Sample bias

> *It raises when data is not a good representation of "whole" data the sample suppose to represent.*

Data sample could be large but still not representative.

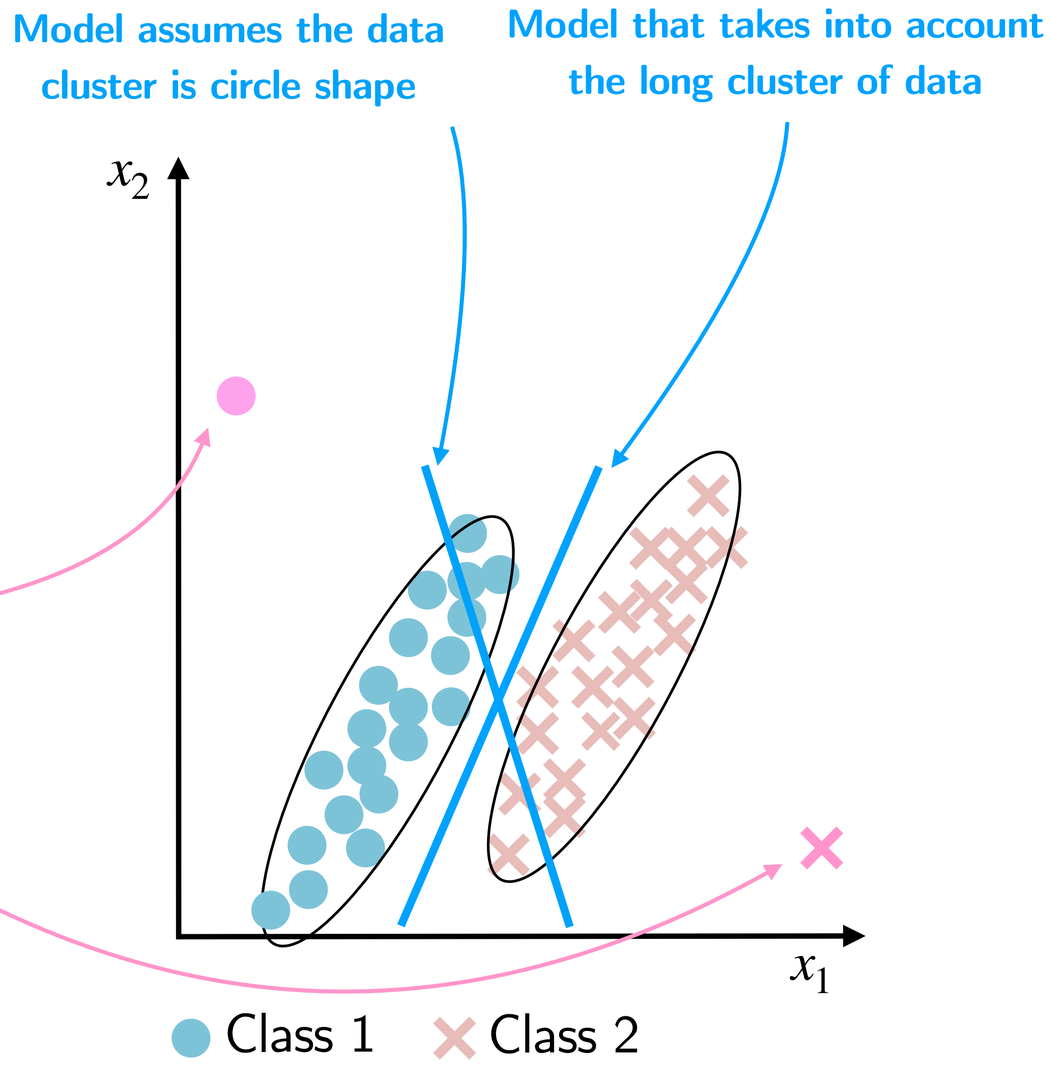
How to Use Inductive Bias and Sample Bias

Inductive bias

Goal of data analyst is to find the correct inductive bias.

Sample bias

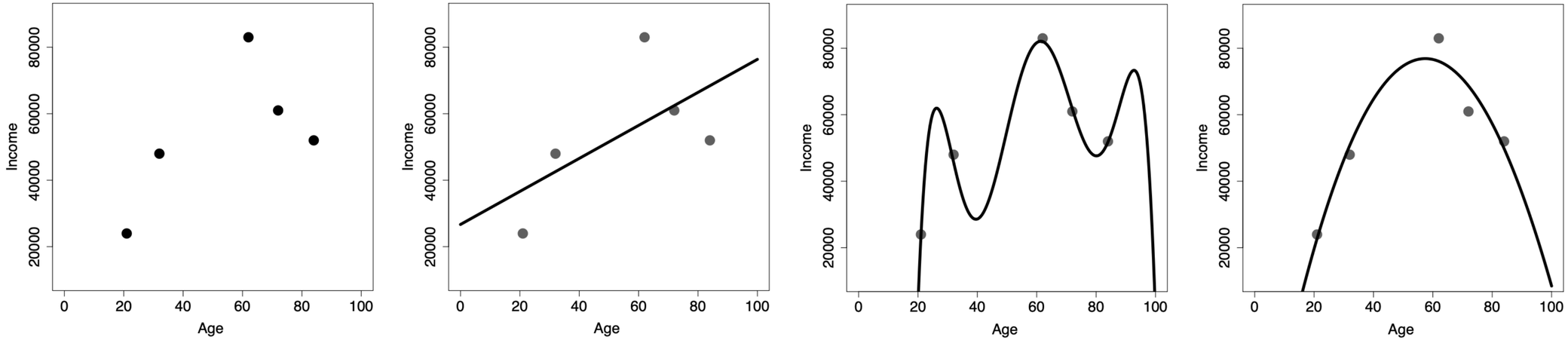
Data analyst should analyze and understand the data to remove this bias.



What Can Go Wrong with the Inductive Bias of Methods?

It can be challenging to know which **inductive bias** works best for a given task or dataset.

The inductive bias (i.e., *model assumptions about the data*) may not be aligned with the data properties (statistics).



(a) Dataset (b) Underfitting (c) Overfitting (d) Just right

figure from ML for Predictive Data Analytics textbook.

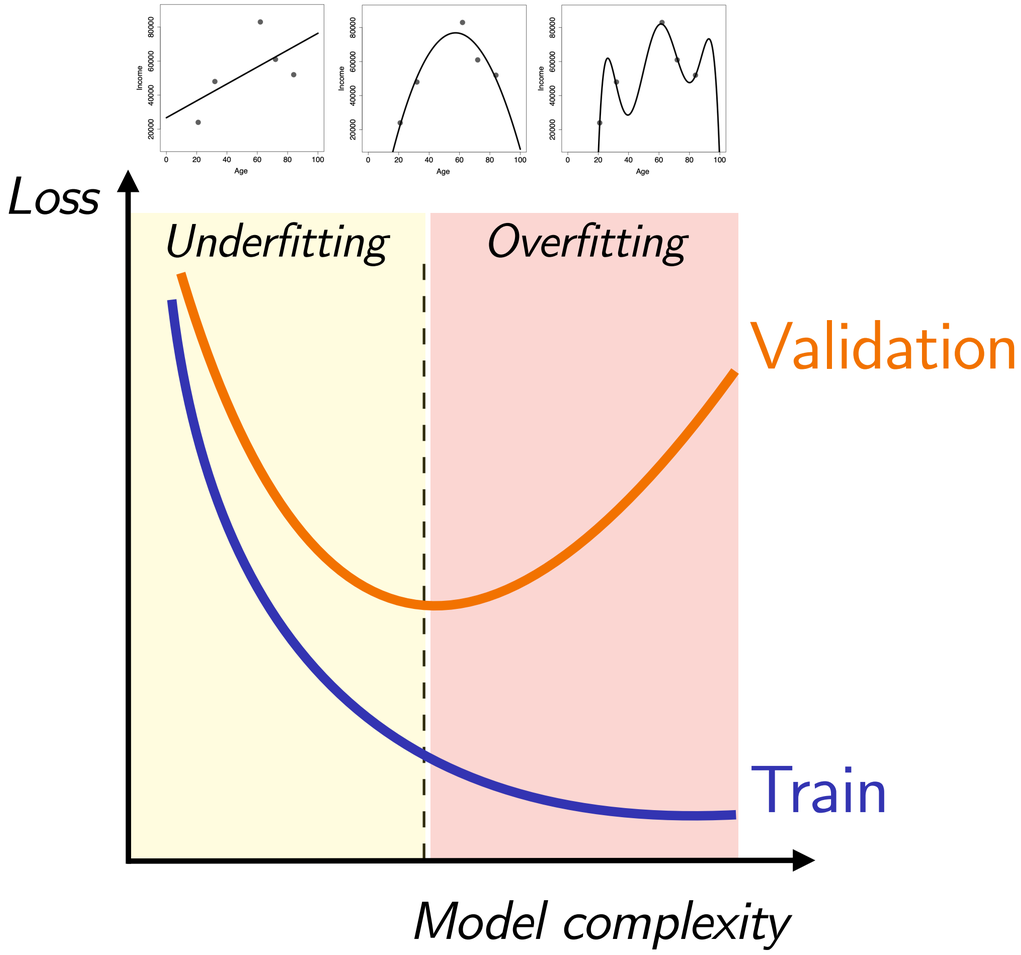
Underfitting and Overfitting, Training/Validation Curves

Underfitting

Model is too simplistic.

Overfitting

Model is too complex
(fitting the train data too well).



Number of train iteration in neural networks

Lifecycle

Cross Industry Standard Process for Data Mining (CRISP-DM)

- A process used to mange project through its lifecycle.
- It is application, industry, and tool natural.
- It views the analytics process from both **application** and **technical** perspectives.

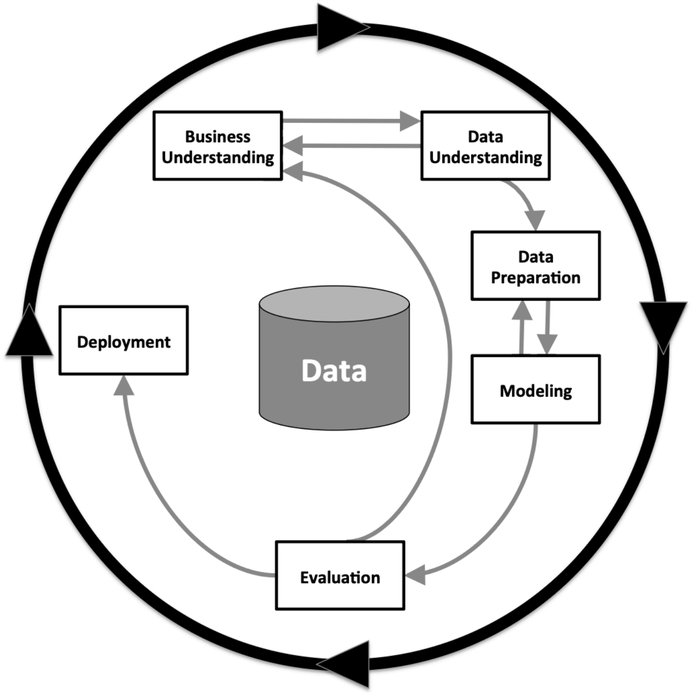


figure from ML for Predictive Data Analytics textbook.

Cross Industry Standard Process for Data Mining

1. To fully understand a business problem

2. How it will be used to address a business problem

3. What kind of data are required (from multiple sources)

4. Machine learning at work

5. Full evaluation and proof

6. Integration of model with processes in organization

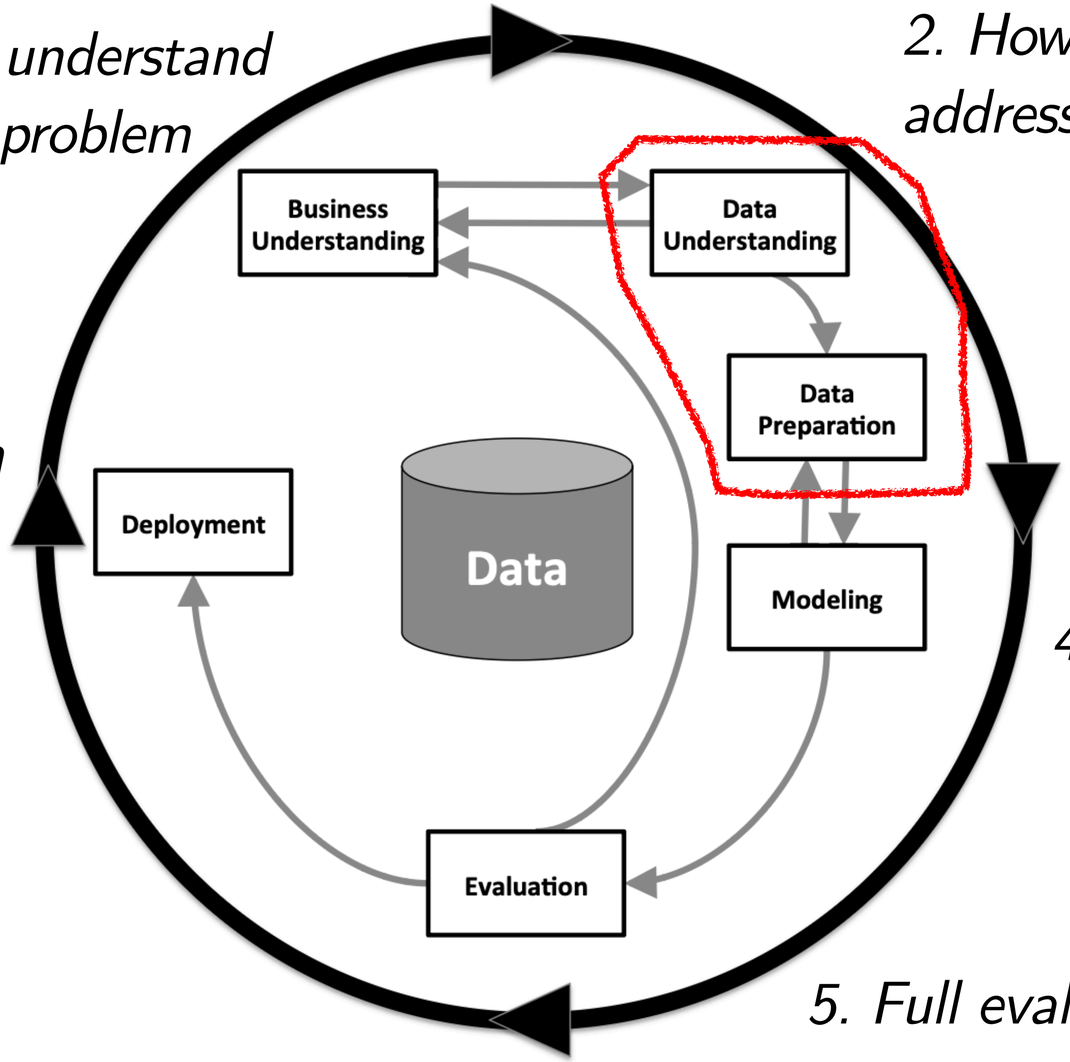


figure from ML for Predictive Data Analytics textbook.

Data Exploration - Getting to Know Your Data

Data Characteristics

Understanding the data, i.e., type of features, their ranges, and distribution of values

Data Quality

Evaluate the quality of your data, i.e., missing values and outliers

— *ML for Predictive Data Analytics*

Data Characteristics - Statistics

To learn about data

- Central tendency (mean, mode, median)
- Variation (standard deviation, percentiles)
- **Visualization** (bar plots, histogram, box plots)

Visualization

Visualization

Visualization

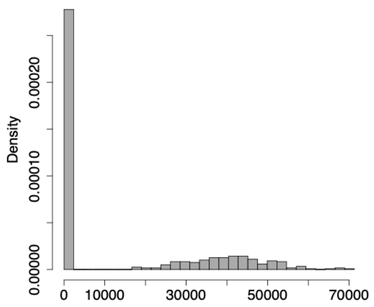
Visualization

Visualization

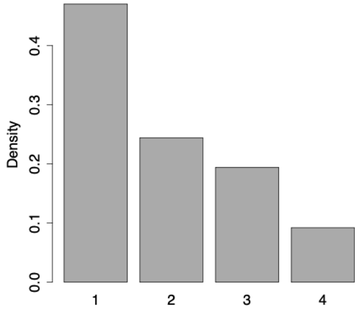
Visualization

Visualization

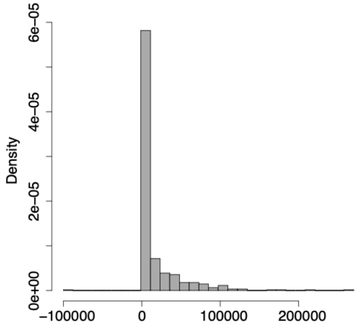
Data Visualization (Ex. Insurance Related Data)



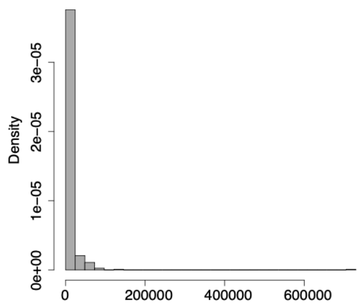
(a) Income



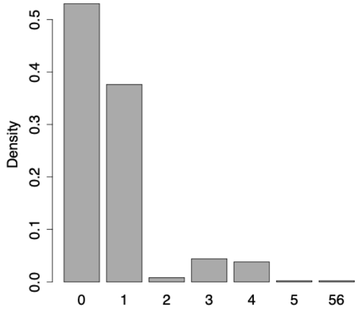
(b) Num. Claimants



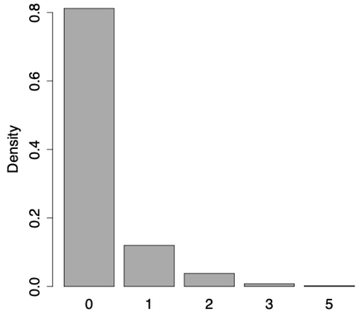
(c) Claim Amount



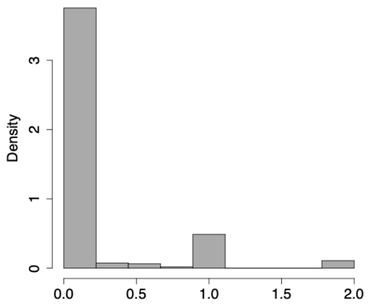
(d) Total Claimed



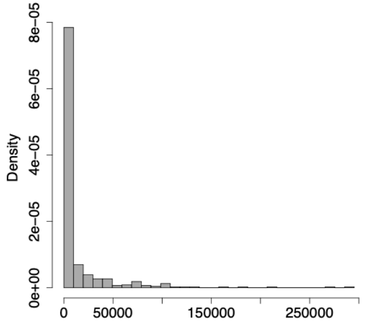
(e) Num. Claims



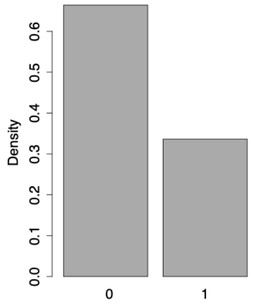
(f) Num. Soft Tissue



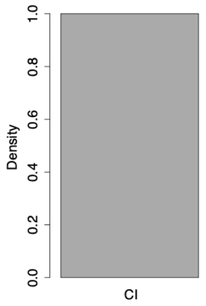
(g) % Soft Tissue



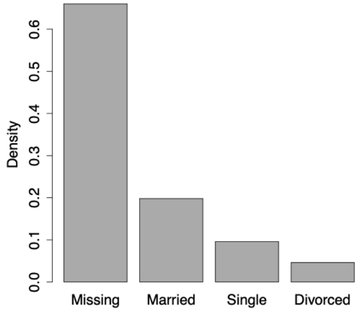
(h) Amount Received



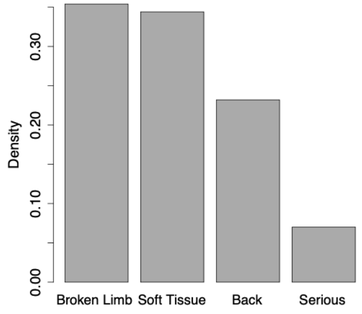
(i) Fraud Flag



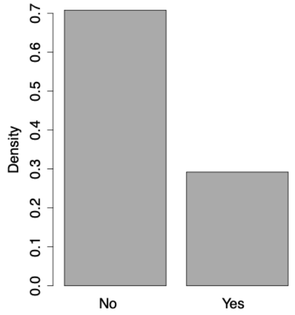
(j) Insurance Type



(k) Marital Status



(l) Injury Type



(m) Hospital Stay

figure from ML for Predictive Data Analytics textbook.

Common Data Distributions

Histograms: A visual representation of the data distribution

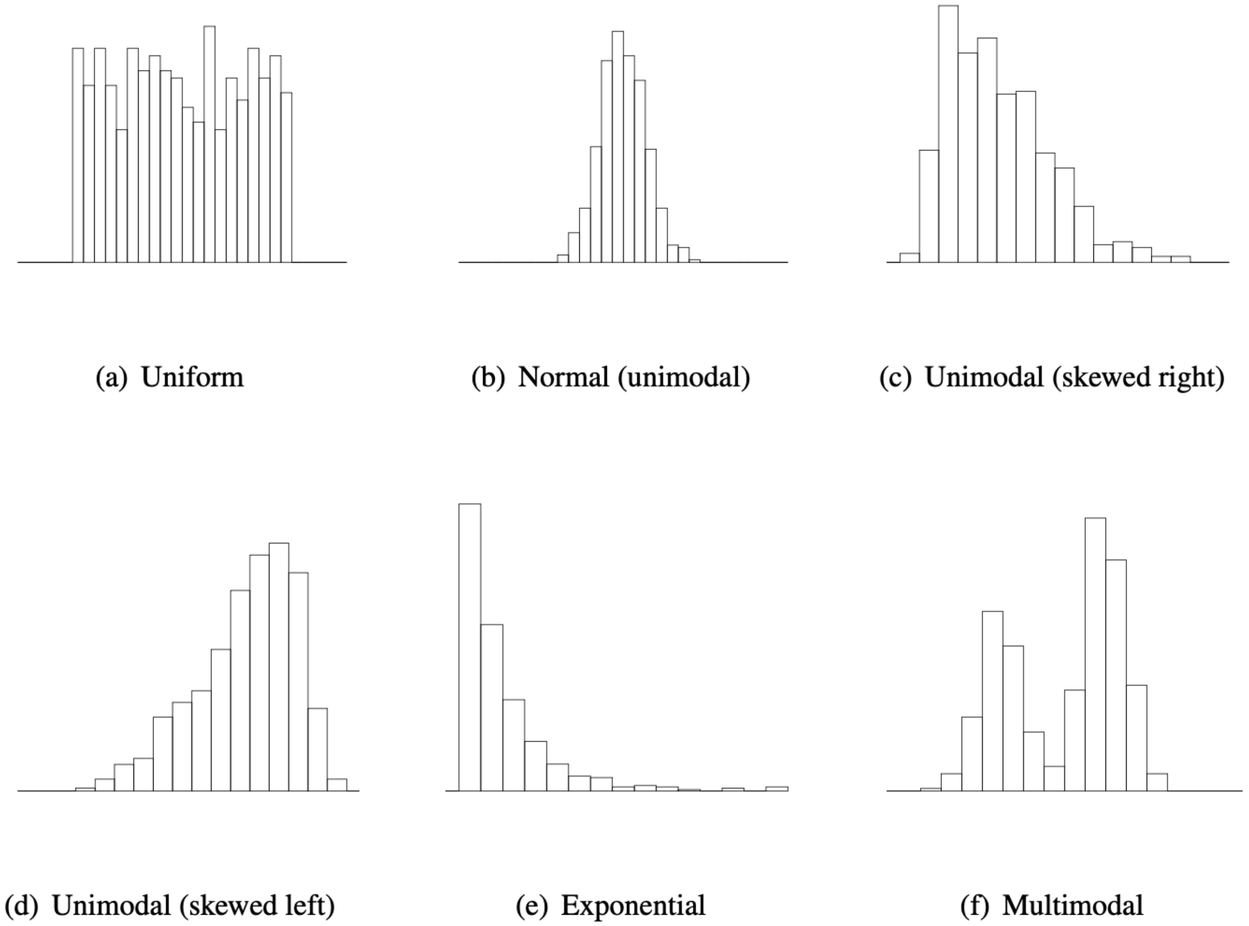
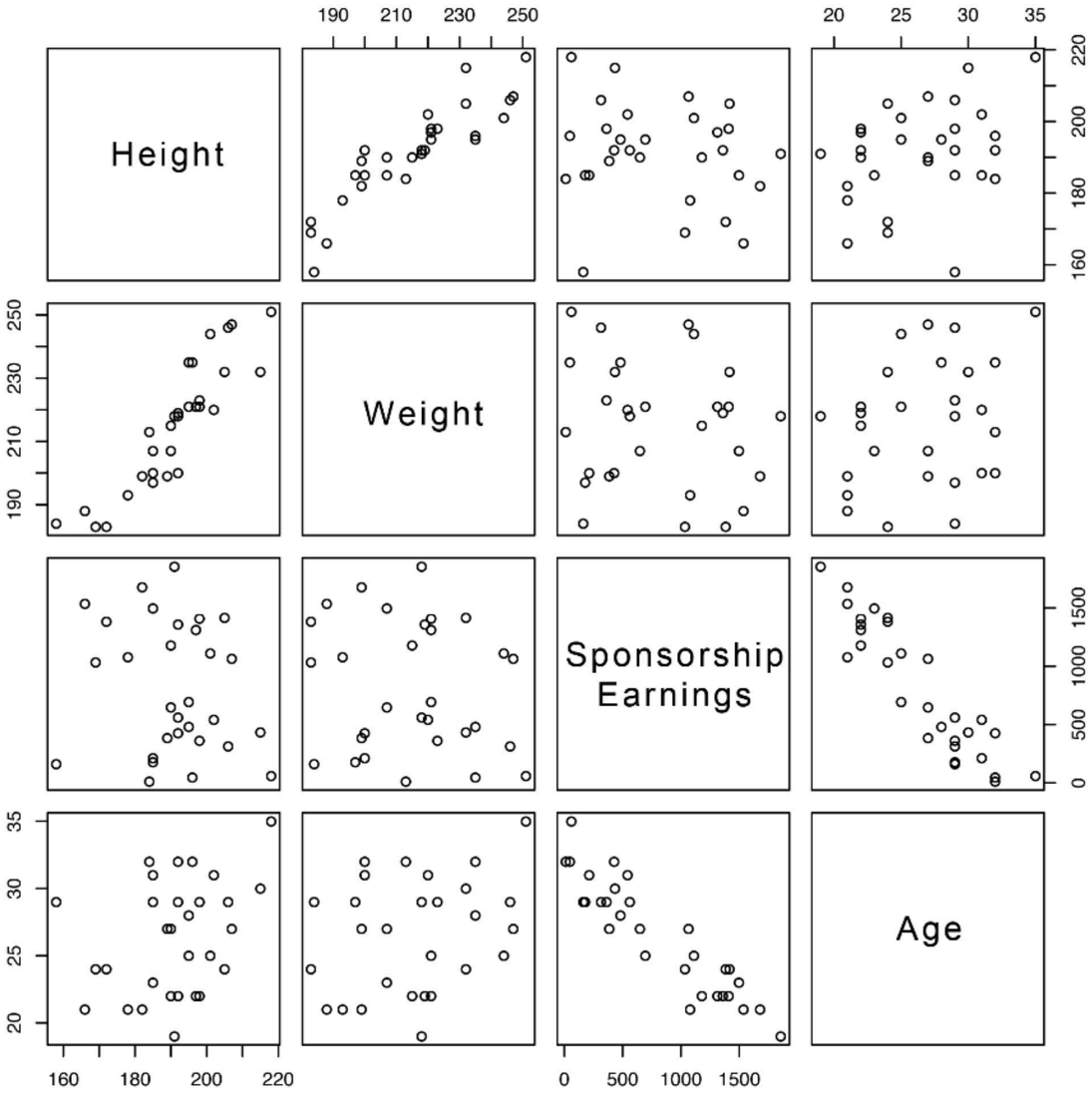


figure from ML for Predictive Data Analytics textbook.

Visualization Technique for Continuous Features (scatter plots)

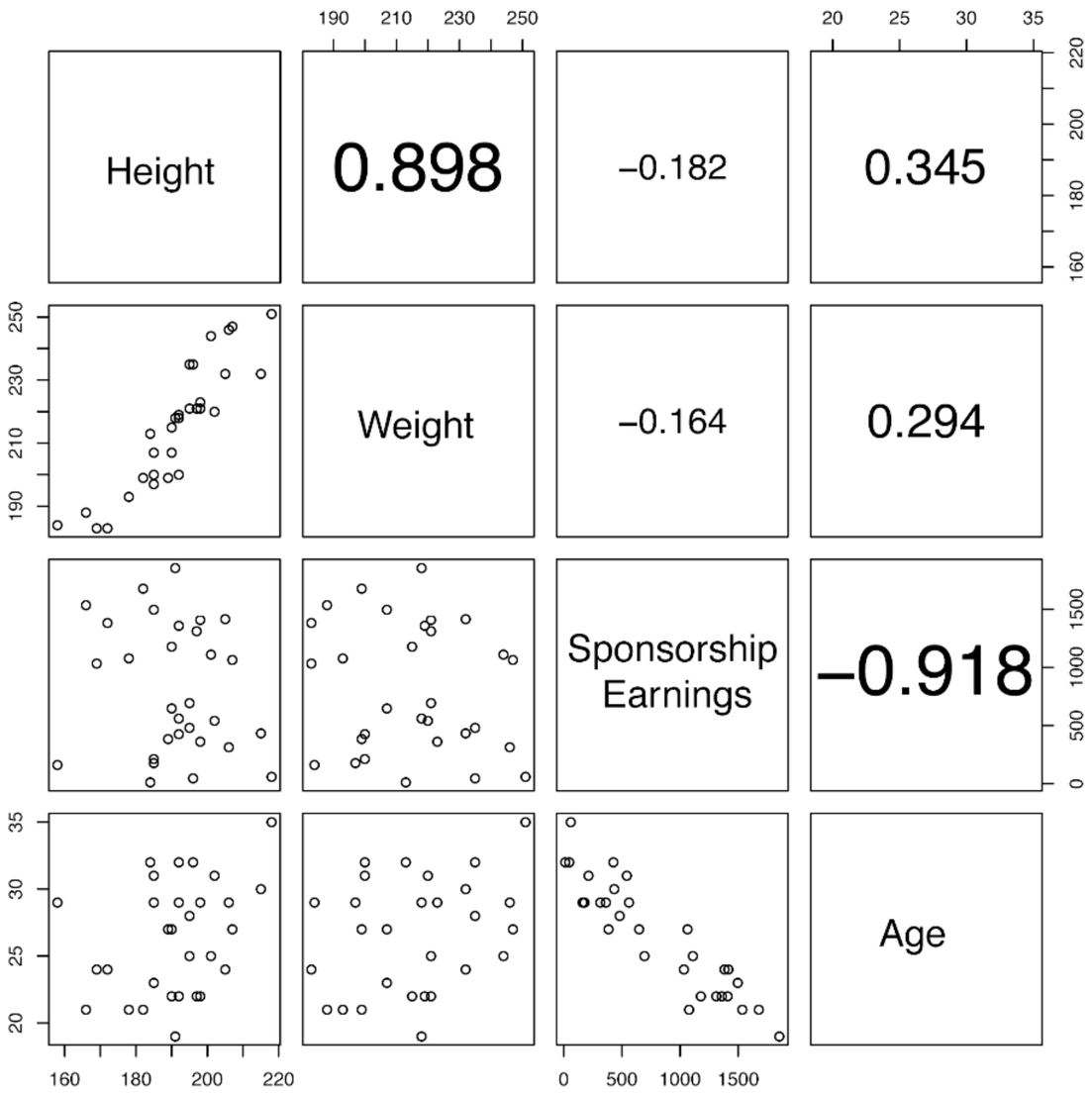


ID	Position	Height	Weight	Career Stage	Age	Sponsorship Earnings	Shoe Sponsor
1	forward	192	218	veteran	29	561	yes
2	center	218	251	mid-career	35	60	no
3	forward	197	221	rookie	22	1,312	no
4	forward	192	219	rookie	22	1,359	no
5	forward	198	223	veteran	29	362	yes
6	guard	166	188	rookie	21	1,536	yes
7	forward	195	221	veteran	25	694	no
8	guard	182	199	rookie	21	1,678	yes
9	guard	189	199	mid-career	27	385	yes
10	forward	205	232	rookie	24	1,416	no
11	center	206	246	mid-career	29	314	no
12	guard	185	207	rookie	23	1,497	yes
13	guard	172	183	rookie	24	1,383	yes
14	guard	169	183	rookie	24	1,034	yes
15	guard	185	197	mid-career	29	178	yes
16	forward	215	232	mid-career	30	434	no
17	guard	158	184	veteran	29	162	yes
18	guard	190	207	mid-career	27	648	yes
19	center	195	235	mid-career	28	481	no
20	guard	192	200	mid-career	32	427	yes
21	forward	202	220	mid-career	31	542	no
22	forward	184	213	mid-career	32	12	no
23	forward	190	215	rookie	22	1,179	no
24	guard	178	193	rookie	21	1,078	no
25	guard	185	200	mid-career	31	213	yes
26	forward	191	218	rookie	19	1,855	no
27	center	196	235	veteran	32	47	no
28	forward	198	221	rookie	22	1,409	no
29	center	207	247	veteran	27	1,065	no
30	center	201	244	mid-career	25	1,111	yes

Basketball data

figure from ML for Predictive Data Analytics textbook.

Visualization of the Correlation Coefficients

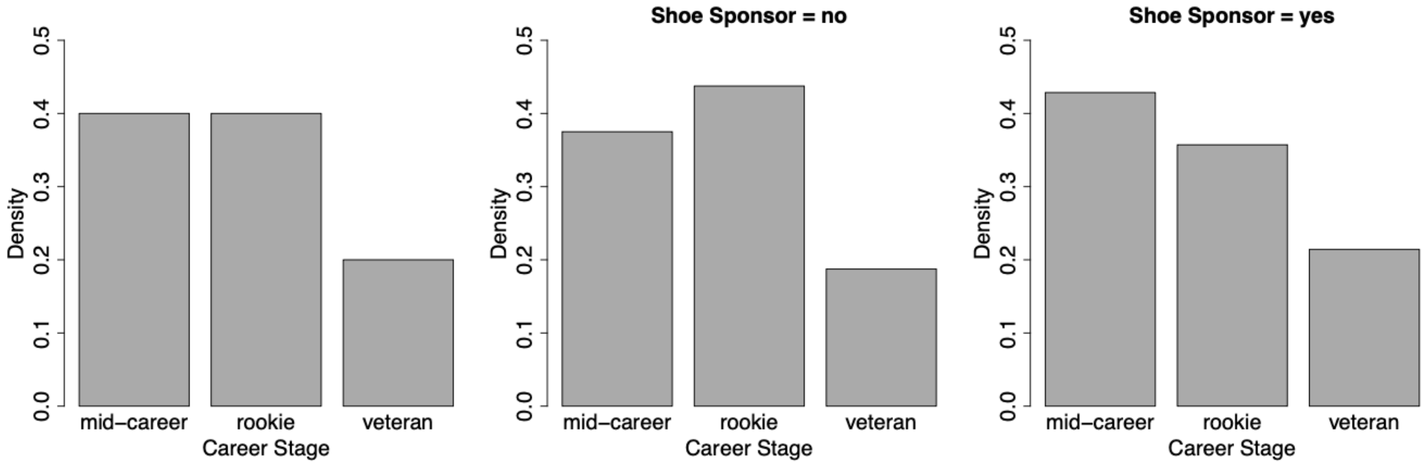


Basketball data

figure from ML for Predictive Data Analytics textbook.

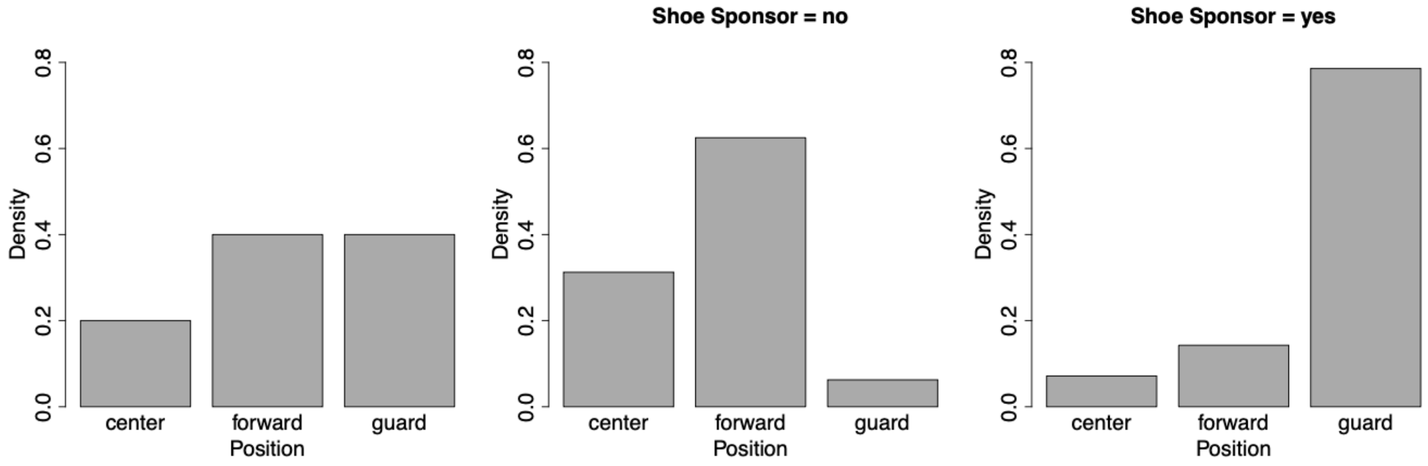
Visualization Technique for Categorical Features (small multiples)

The relationship between:
Career Stage and *Sponsor* features



(a) Career Stage and Shoe Sponsor

The relationship between:
Position and *Sponsor* features



Basketball data
figure from ML for Predictive Data Analytics textbook.

Data Quality

Common challenges

- Missing values
  - Removal or imputation of the feature.
- Irregular cardinality
  - e.g., Cardinality $= 1$, i.e., that all data has similar values for that feature.
  - e.g., Cardinality $= 2$, i.e., it might be a categorical feature.
  - e.g., One feature might have much larger cardinality than others.
- Outliers
  - Invalid due to errors.
  - Valid (nature of data), suggest to do *clamp transformation*.

Clamp Transformation to Handle Outliers

Clip *very large* and *very small* feature values

$$a_i = \begin{cases} a_{\text{lower}}, & \text{if } a_i < a_{\text{lower}} \\ a_{\text{upper}}, & \text{if } a_i > a_{\text{upper}} \\ a_i, & \text{otherwise} \end{cases}$$

where a_{lower} and a_{upper} can be set manually based on *domain knowledge* or can be calculated from data (next slide).

Two Approaches to Compute the Clamp Range from Data

We are trying to compute how large/small is too large/small!

Inter-quartile Approach

- a_{lower}: 1st quartile value minus 1.5 times inter-quartile range.
- a_{upper}: 3rd quartile value plus 1.5 times inter-quartile range.

where Inter-quartile range is the difference between 25th percentile and the 75th percentile.

1st (lower) quartile

3rd (upper) quartile

Normal Distributed Low Probability

- a_{lower}: mean value minus 2 times the standard deviation.
- a_{upper}: mean value plus 2 times the standard deviation.

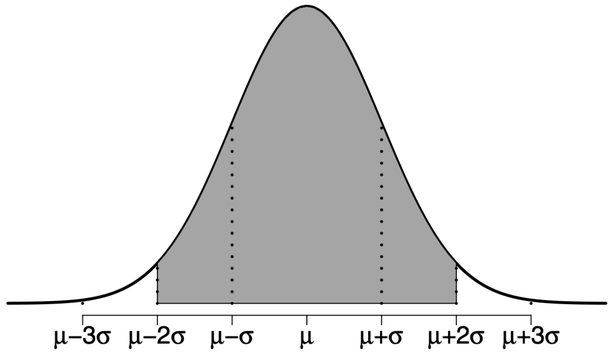


figure from ML for Predictive Data Analytics textbook.

What Have We Done So Far?

- Data exploration
  - Visualize data
  - Understand its range, cardinality, and relation
- Data quality
  - Remove or clamp feature that you find problematic
- Data preparation
  - Pre-process the data

Data Preparation

To make the data more compatible with certain machine learning methods, or to handle noise within the data.

Two common techniques:

- Normalization and standardization
- Binning

Data Preparation — Normalization

To mitigate/eliminate a problem of dealing with different ranges of data (each feature different range).

Normalization

"changes" a continuous feature to fall into a specific range [low, high].

$$a_i^{\text{normalized}} = \frac{a_i - \min(a)}{\max(a) - \min(a)} * (\text{high} - \text{low}) + \text{low}$$

Drawback: sensitive to outliers!
Possible solution? —> next slide.

Data Preparation — Standardization

Standardization

Shifts and scales the feature by its mean and standard deviation.

$$a_i^{\text{standardized}} = \frac{a_i - a_{\text{mean}}}{a_{\text{standard deviation}}}$$

Majority of values will be in $[-1,1]$ (not all though).

It assumes data is normally distributed.

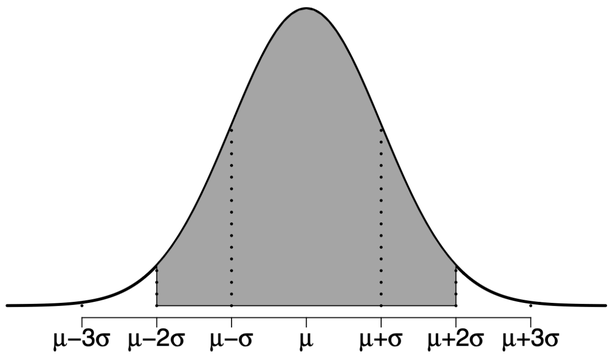


figure from ML for Predictive Data Analytics textbook.

Should we do **normalization** or **standardization**?

1 Go to wooclap.com

2 Enter the event code in the top banner

Event code
OJCYPS

Enable answers by SMS

Normalization vs. Standardization

There is no hard and fast rule that says which is the best normalization technique, and this decision is generally made based on experimentation.

— *ML for Predictive Data Analytics*

Some machine learning methods may have certain assumptions on the data (Gaussian distributed, or be within a particular range). Hence, insights about each method can be helpful to know how to prepare the data.

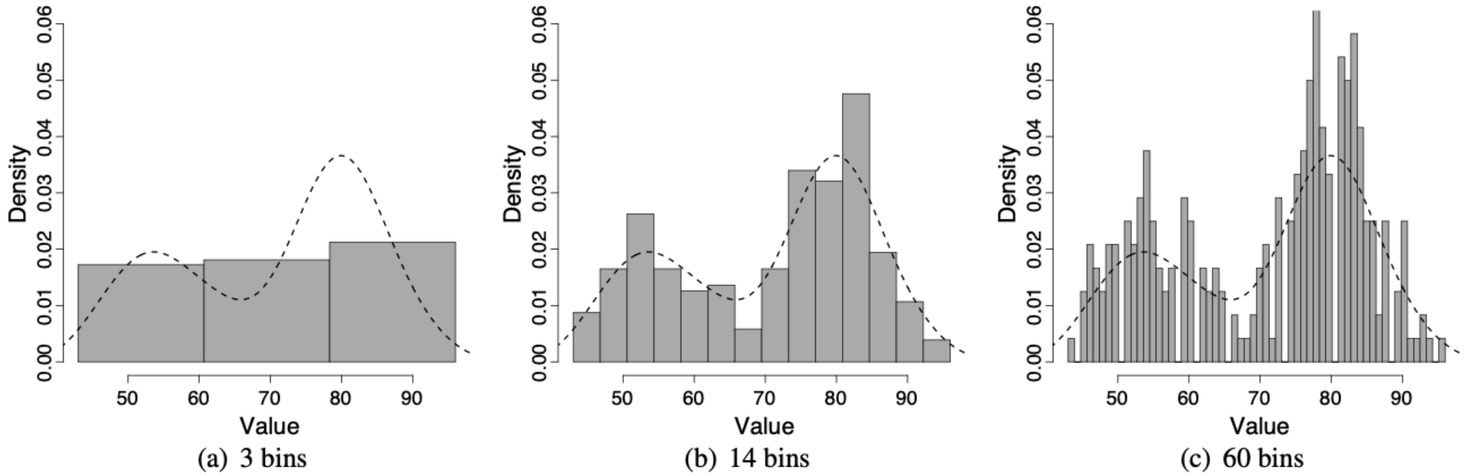
— *Bahareh*

Data Preparation — Binning

To convert continuous features into a categorical feature.

Binning

*Define a series of ranges **(bins)** for the continuous feature that correspond to the levels of the new categorical feature we are creating.*



How to choose the bins?
How many bins?

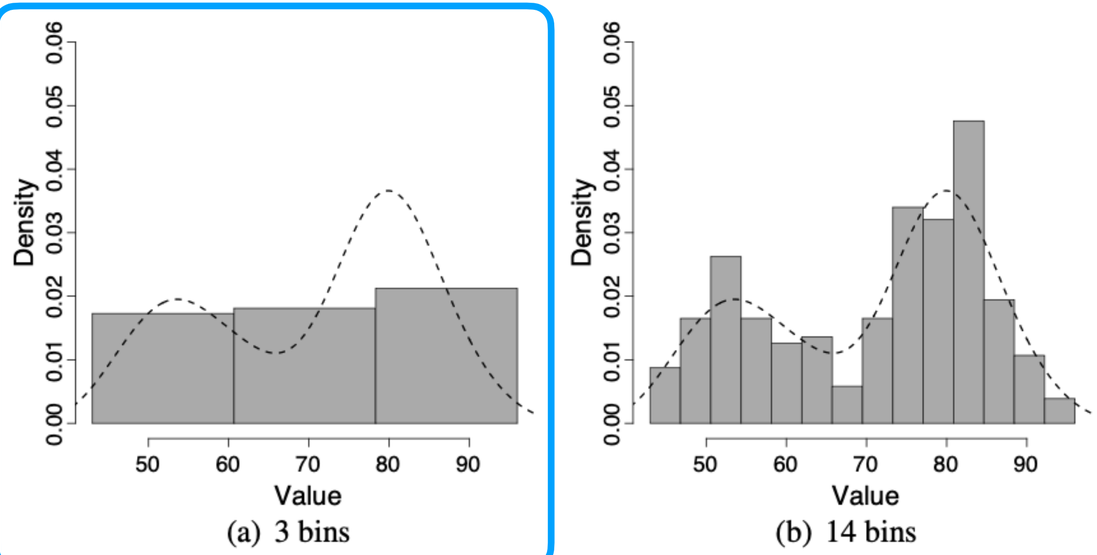
figure from ML for Predictive Data Analytics textbook.

How to Choose the Number of Bins?

This is why visualization and understanding your data is important.

Intuition and a process of trial and error experimentation is needed.

Challenges



If number of bins is very low (2 or 3), lots of information can be lost (but it means a lot of values in each bin).

If number is very high, higher possibility of "finding" interesting, better representation of feature distribution, yet we could have empty bins.

figure from ML for Predictive Data Analytics textbook.

Two Approaches to Bin the Data

Equal-width Binning (b bins):

$$\frac{\text{range of data}}{b}$$

Simple, intuitive, yet if distribution is "far" from uniform, bins end up with very different number of values.

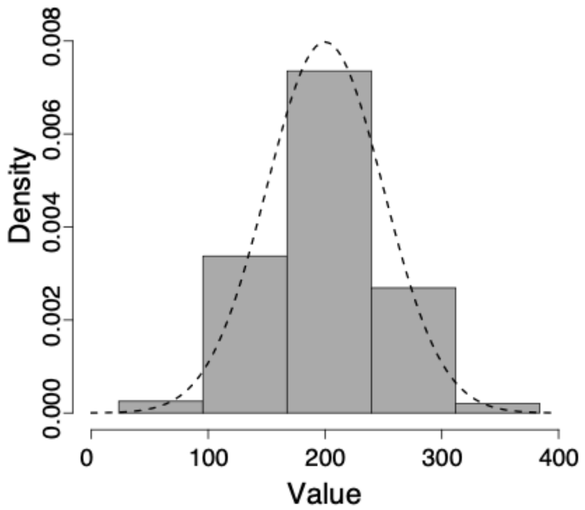
Equal-frequency Binning (b bins):

$$\frac{\text{number of values}}{b}$$

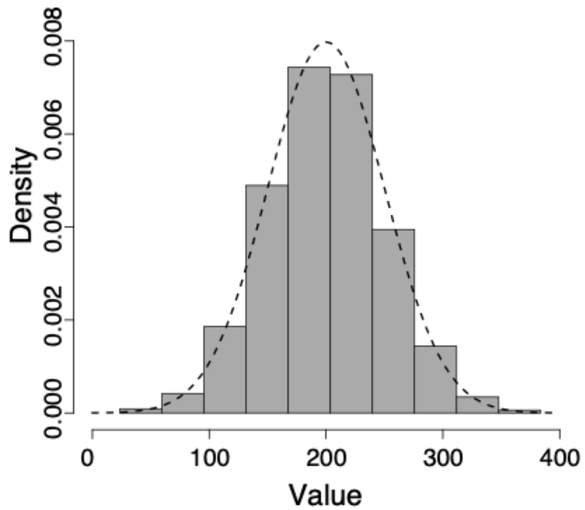
Sort values into ascending order, then place equal number of values into each bin.

Visualized

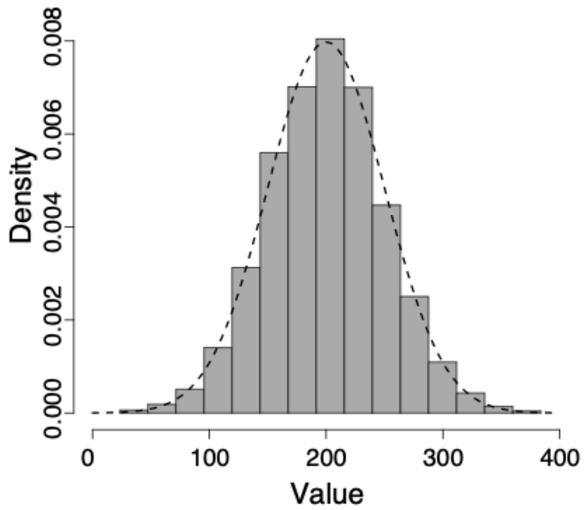
Equal-width Binning



(a) 5 equal-width bins

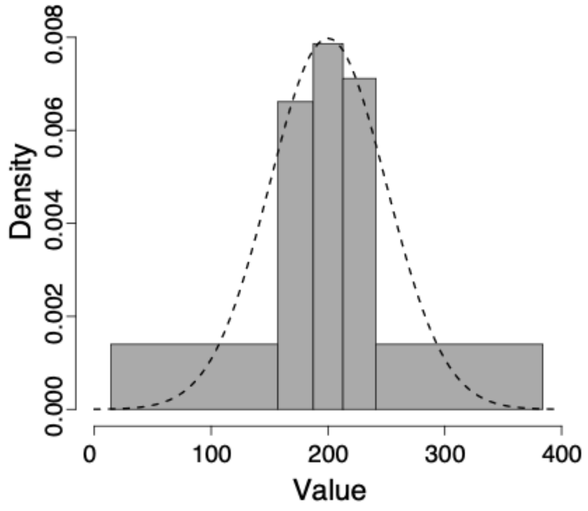


(b) 10 equal-width bins

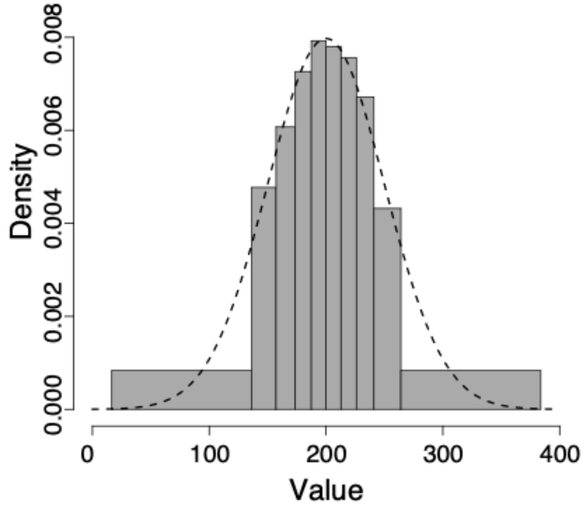


(c) 15 equal-width bins

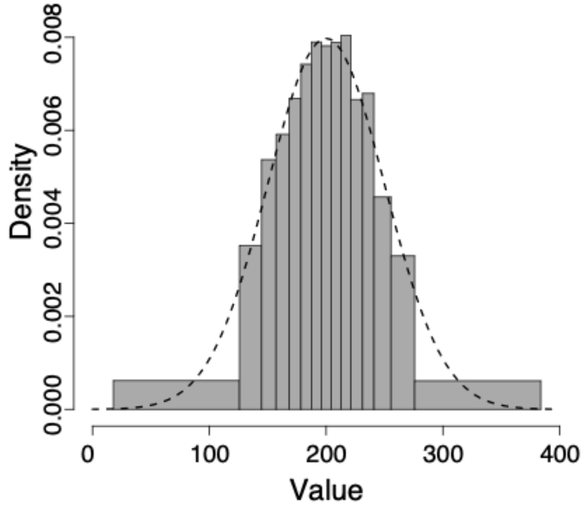
Equal-frequency Binning



(d) 5 equal-frequency bins



(e) 10 equal-frequency bins



(f) 15 equal-frequency bins

What If the Data is too Large? — We **Sample** It

Sample a smaller percentage from the larger dataset.

- Done carefully so the sample is representative.
- Distribution of the sample resembles/same as distribution of data.

Two approaches

- Top sampling
- Random sampling

Top Sampling

AVOID top sampling!

We often have a data frame (table or vector of datasets). One may extract the first s out of many available data samples.

Top Sampling AVOID top sampling!

selects the top $s\,\%$ of instances from a dataset to create a sample.

This runs a serious risk of introducing bias. The samples are affected by any ordering of the original dataset.

AVOID top sampling!

Recommended Technique for Sampling

Random Sampling

Randomly select a proportion of $s\,\%$ of the instances from a dataset to create a sample.

You are sampling uniformly at random.

How to Sample Categorical Target Data?

Goal: Make sure to preserve the distribution of target categories.

If there are one or more levels of a categorical feature that only a very small proportion of instances in a dataset have, there is a chance that these will be omitted or underrepresented by random sampling.

We should ensure that the relative frequencies of the level of a specific stratification feature are maintained in the sampled dataset.

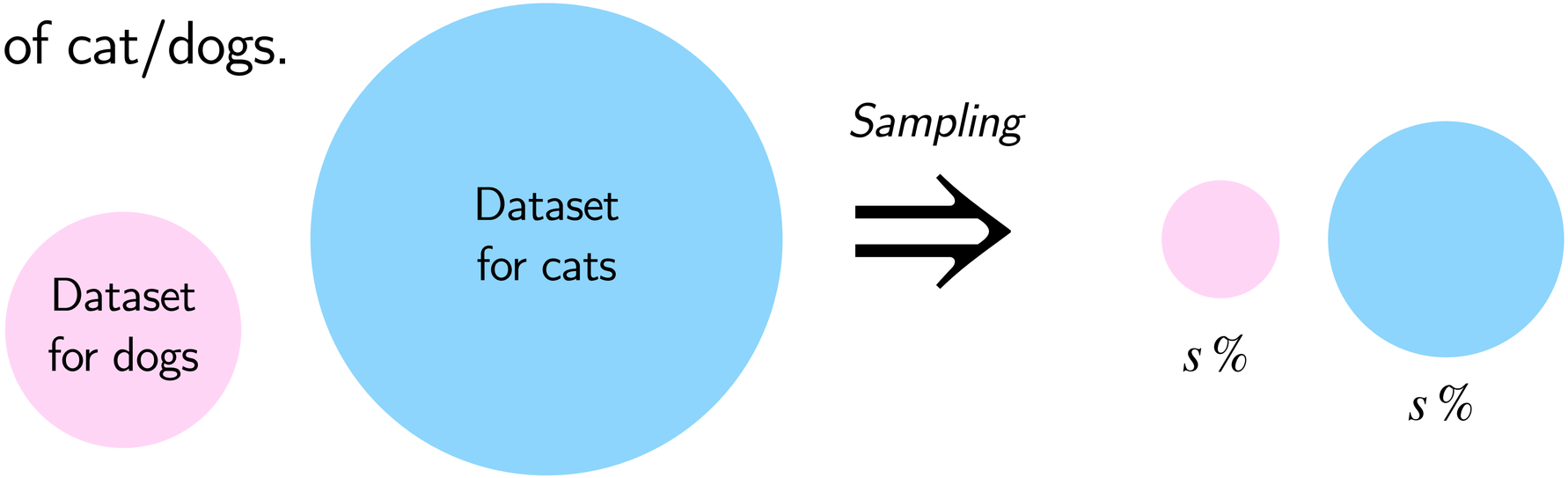
— *ML for Predictive Data Analytics*

Stratified Sampling for Categorical Target Data

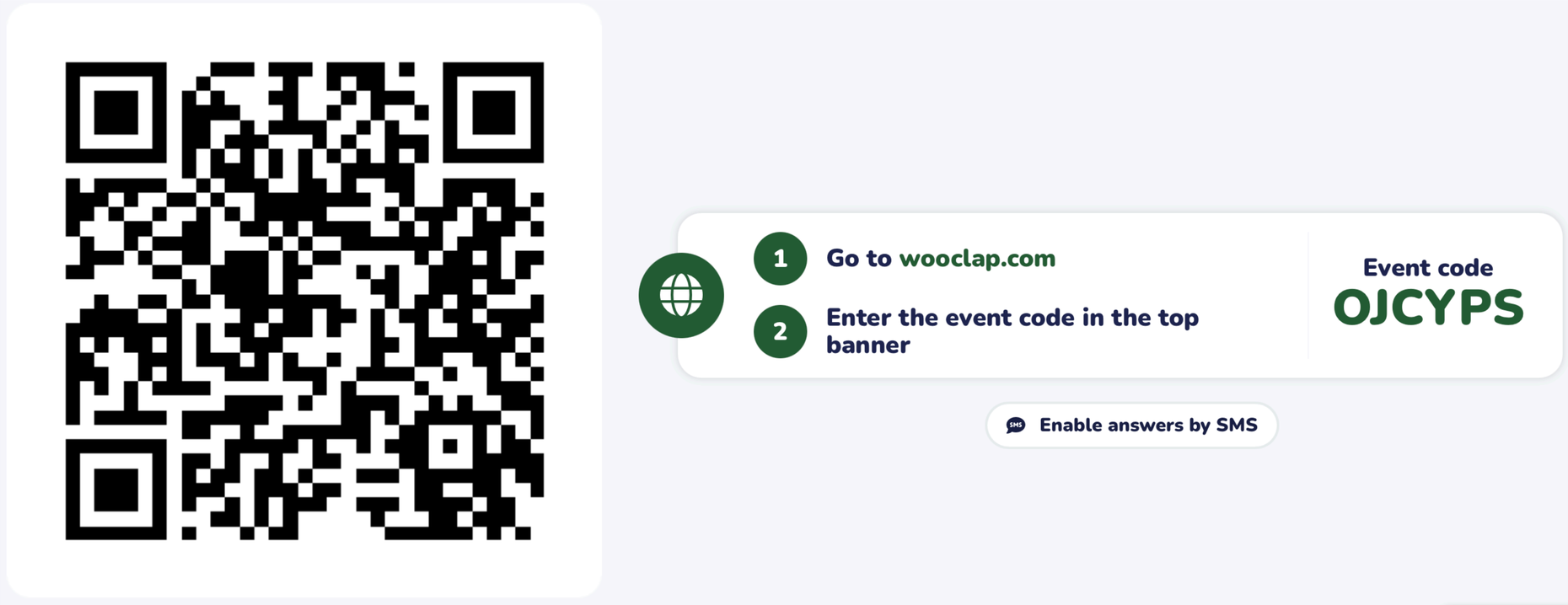
Stratified Sampling

1. *Divide data into groups (strata), each group one value of feature.*
2. *Take $s\,\%$ from each group/stratum.*
3. *Combine all of them ($s\,\%$ data points).*

Ex. Classification of cat/dogs.



Imagine you have trained a classifier on your (98 cats, 2 dogs) image dataset; and its learned logic is to classify any input image as a cat. What is the accuracy of this classifier on a test set with (98 images of cats, and 2 images of dogs)?



Challenges of Non-Equal Distribution of Target Data

Imbalanced Dataset

Not equal distribution of target features

e.g., 98 images of cats, 2 images of dogs.

Imagine you have trained a classifier on your (98 cats, 2 dogs) image dataset; and its learned logic is to classify any input image as a cat. What is the accuracy of this classifier on a test set with (98 images of cats, and 2 images of dogs)?

98%

Imbalanced data often trick naive machine learning engineers! Be careful!

How to deal with imbalanced data? (next slide)

Create A Sample with Equally Represented Categorical Features

Under-sampling

Divide data into groups: no change for <u>smallest</u> group, and other groups are randomly sampled to the size of the smallest group.

Over-sampling

Divide data into groups: no change for <u>largest</u> group, and other groups randomly sampled with replacement to the size of the largest group.

Takeaways (I)

- ***Machine learning process*** — *Training (model building) followed by its utilization.*

- ***Inductive bias*** — *Concerned with selection of type of models and its specifics.*

- ***Sample bias*** — *Concerned with representative data sample.*

- ***Underfitting and overfitting*** — *An art of finding a "right", generalized, model.*

Takeaways (II)

- ***To know your data*** — *Central tendencies, variations, distributions.*

- ***Inductive bias*** — *Concerned with selection of type of models and its specifics.*

IMPORTANT

- ***Keep "true" characteristics of data***
- ***Keep track of any processing steps*** — *Anything performed on training/testing data has to be done of "real" data when the model is deployed.*

Amii?

Alberta Machine Intelligence Institute (Amii)